TECHNICAL REPORT

MDII Scoring Dashboard (Version 1.00): A Tool for Visualizing Digital Inclusiveness and Innovation Performance

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ABSTRACT

The MDII scoring dashboard was developed as part of a set of tools to assess the social inclusiveness of digital innovations across food, land, and water systems. This tool provides decision-makers with interactive visualizations and key metrics such as accessibility, user engagement, and cultural sensitivity. The dashboard, shaped by collaborative workshops and feedback, incorporates metadata indicators, feedback mechanisms, and customizable data views for portfolio managers. By integrating both quantitative and qualitative data, it allows for monitoring performance, identifying gaps, and aligning digital tools with inclusiveness goals, supporting adaptive learning and sustainability in digital innovations.

SUMMARY

The MDII Scoring Dashboard is an innovative tool designed to provide critical insights into the inclusiveness and impact of digital tools used within agri-food systems. By breaking down performance across seven key dimensions—Accessibility, Beneficial Impact, Usage Effectiveness, Ethical & Responsible Innovation, Co-Creation & Governance, Risks & Harms, and Supportive Ecosystem—the dashboard enables users to evaluate digital innovations from multiple perspectives.

Each dimension features detailed sub-dimensions, metadata, and qualitative feedback from tool beneficiaries, allowing for a nuanced understanding of strengths and areas for improvement. The interactive visuals, including radar charts, waffle charts, and decomposition trees, facilitate deep analyses, while customizable filters enhance flexibility, enabling exploration of data by tool, country, portfolio, and technology.

Moreover, the dashboard also allows focus on ethical and responsible practices, ensuring that digital tools respect community rights and contribute positively to social equity. Sections dedicated to assessing risks and fostering a supportive ecosystem further underscore its comprehensive approach. Ultimately, the MDII Scoring Dashboard serves as a vital resource for portfolio managers, decision-makers, and stakeholders, providing a transparent, data-driven framework for monitoring the inclusiveness, sustainability, and performance of digital innovations.

INTRODUCTION

A dashboard is a visual tool that consolidates and displays key performance indicators (KPIs) and critical data points in a single view. Its primary purpose is to facilitate informed decision-making by presenting data in an easily understandable format. Dashboards help in monitoring performance, identifying trends, and highlighting areas that require attention (Zingde & Shroff, 2020). For instance, Dong et al., (2020) used an interactive dashboard to track and monitor Covid-19 infections in real- time.

The Multi-Dimensional Digital Inclusiveness Index (MDII) (Opola *et al*., 2023; Martins *et al*., 2023) dashboard was developed to provide a comprehensive and interactive tool for assessing the inclusiveness of digital solutions within the Agriculture, Food, Water, and Land (aFWL) sectors. The MDII seeks to ensure that digital tools are not only accessible, usable, and safe but also offer tangible benefit all actors within the aFWL system, with a particular focus on marginalized groups such as women, indigenous communities, and youth in rural areas (Martins *et al*., 2023).

The primary objective of the MDII Scoring Dashboard is to provide decision-makers with a data-driven, user-friendly interface for monitoring and assessing the inclusiveness of digital tools. The dashboard aims to visualize key metrics across user groups and geographies, track the impact of digital tools on underserved populations, and offer insights through customizable reports. Additionally, by incorporating feedback mechanisms, the dashboard supports ongoing improvements in digital inclusiveness and sustainability. Dashboards like the MDII offer industry-wide value by enhancing data accessibility, accelerating decision-making, and fostering a data-centric culture. They support strategic decisions by consolidating data for quicker, more accurate insights (Eberhard, 2021) and improve efficiency by simplifying information, which reduces cognitive load and errors (Paulsen & Lindsay, 2024). Real-time data monitoring enables timely interventions (Frontiers, 2023), while an intuitive interface boosts user engagement (Paulsen & Lindsay, 2024). Dashboards also help organizations establish a data-driven culture by making insights accessible to all stakeholders (Masiello *et al*., 2024).

The main users of the MDII Scoring Dashboard are portfolio managers, decision-makers, researchers, and innovators involved in digital innovation projects. These users can monitor performance metrics, compare digital tools, and make data-driven decisions to enhance inclusiveness and performance in their portfolios. The MDII Dashboard serves as an essential tool for portfolio managers and decision-makers to visualize inclusiveness metrics and track the performance of digital innovations over time.

The MDII Scoring Dashboard offers several key features:

System Type Analysis: Determines whether a digital tool focuses on food, water, or land systems.

Interactive Visuals: Engaging data representations such as gauge charts, radar charts, waffle charts, and decomposition trees.

Filters and Search: Enables users to customize their view by filtering based on tool, region, or portfolio.

Dimension Breakdown: In-depth analysis of each of the seven dimensions to provide clear insights into tool performance.

The dashboard's unique design and functionality align with the MDII's goal of fostering inclusiveness within digital innovations, providing both quantitative and qualitative insights to aid in promoting equitable solutions. By focusing on the needs of marginalized communities, the dashboard ensures that digital tools not only address technical challenges but also contribute to broader social equity.

COLLABORATIVE EFFORT ON DEVELOPMENT

The dashboard was designed with a human-centered approach, focusing on collaboration with portfolio managers and key stakeholders. This approach prioritized an intuitive and user-friendly experience, allowing alignment with actual needs of its users. By engaging directly with portfolio managers during the design process, we were able to gather valuable insights into their priorities, challenges, and expectations. This collaborative effort informed the dashboard's user-friendly layout, customizable features, and interactive visualizations, making it a practical tool for decision-makers to monitor digital inclusiveness across diverse portfolios. There result is a comprehensive and intuitive dashboard that enables users from diverse backgrounds to interact with the data effectively, fostering meaningful engagement and informed decision-making.

On August 27th, 2024, a collaborative working session was held to refine and align the MDII Scoring Dashboard with the needs of its users. During this session (see Figure 1 and 2), key stakeholders emphasized the importance of adaptive learning and sustainability, suggesting that the dashboard should include features to track how feedback and insights are integrated into future developments. They highlighted the value of having customizable data views to explore specific metrics, such as demographic reach, stakeholder feedback, and the inclusiveness of each digital tool, as reflected on Figure 2. This adaptability aims to support portfolio managers and decision-makers in gaining targeted insights and aligning their analyses with strategic goals. The discussions led to a consensus on the essential features and requirements for the dashboard. It was agreed that the tool should incorporate comprehensive inclusiveness metrics, beyond the MDII scores, to encompass metadata indicators like accessibility, user engagement, and cultural sensitivity. The participants also stressed the need for visual and interactive data presentations, with options to filter and customize views based on user needs, including demographic segmentation and tool feedback. A recurring theme was the balance between quantitative and qualitative data to provide a holistic view of digital inclusiveness.

Finally, there were a few points where opinions varied, particularly regarding the balance between qualitative and quantitative data and the prioritization of learning insights versus development tracking. While some stakeholders prioritized capturing feedback and lessons learned, others emphasized tracking the development progress and tool maintenance. These insights from the working session played a critical role in shaping the dashboard's features and guiding future refinements.

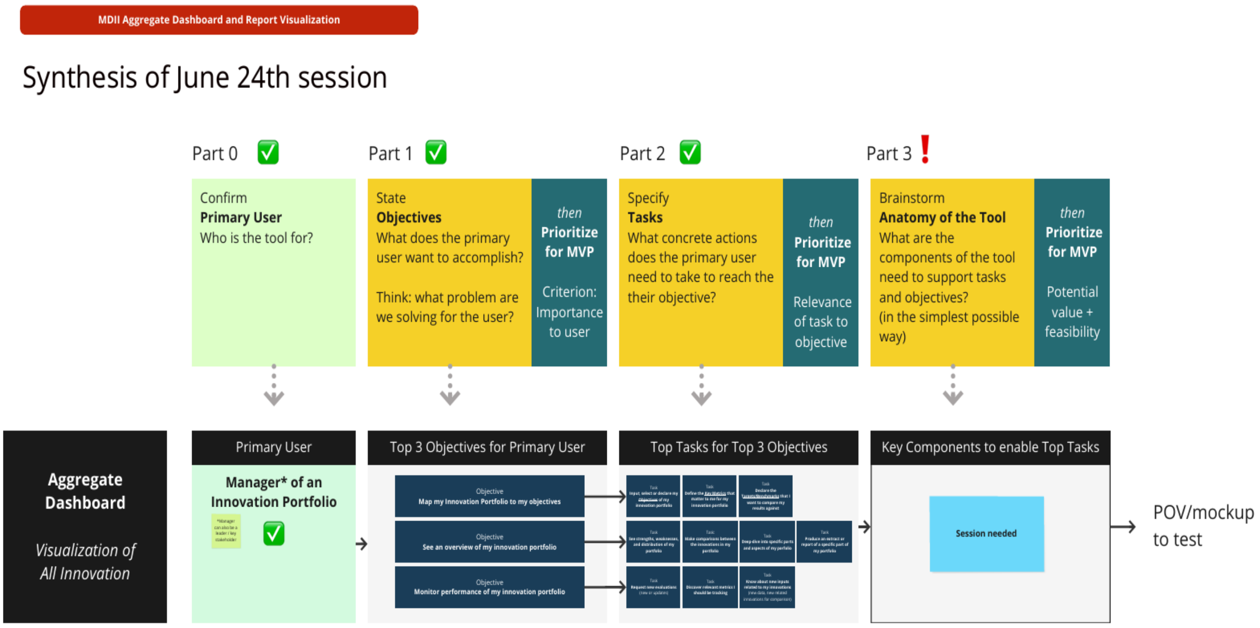

Figure 1: Collaborative session to identify the primary users, objectives and the tasks of the dashboard

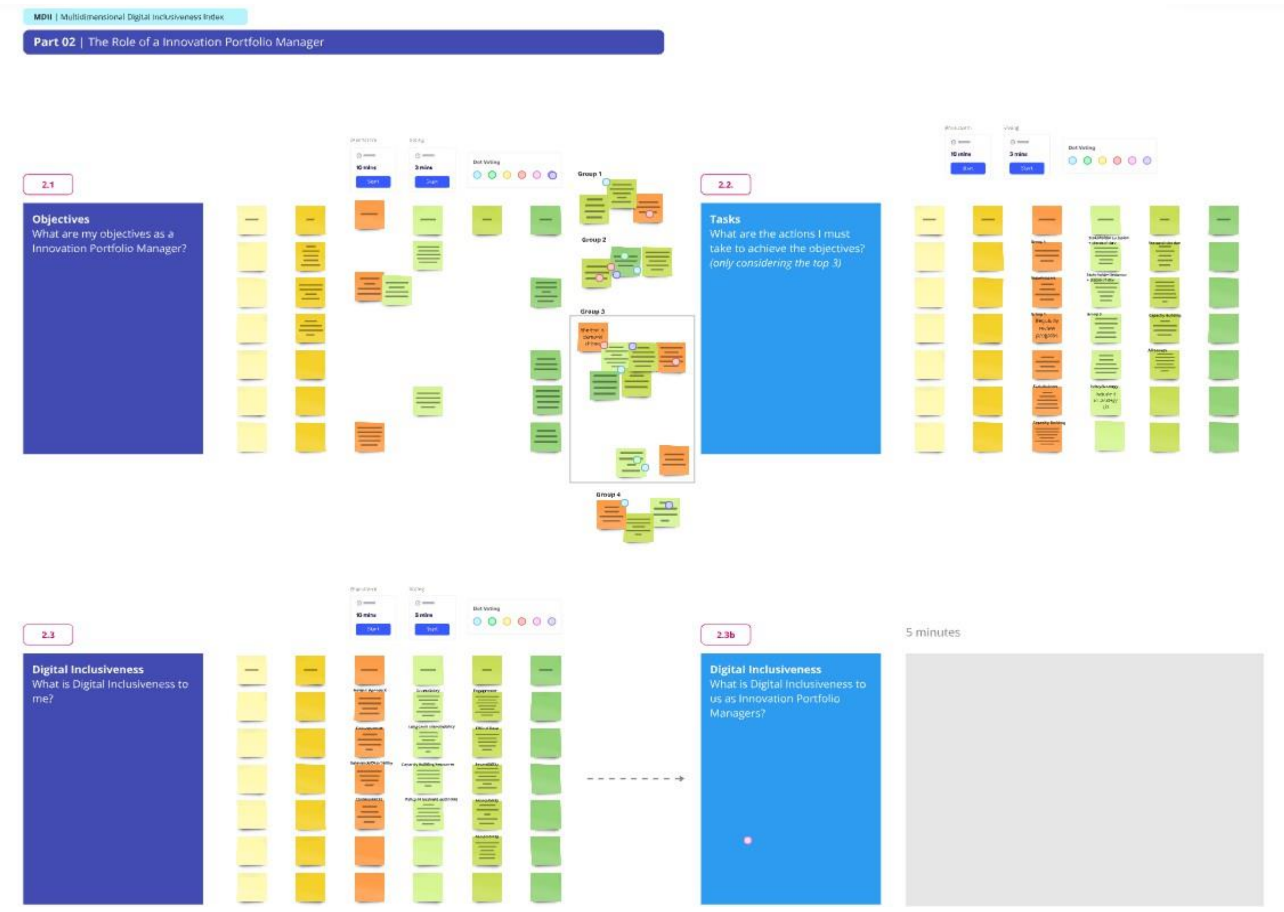

Figure 2: Collaborative Miro board session held on August 27th, 2024, to define and align the objectives, tasks, and understanding of digital inclusiveness for the MDII Scoring Dashboard.

NAVIGATING THE DASHBOARD

The MDII Scoring Dashboard is designed to provide a clear, accessible layout that enables portfolio managers, researchers, and decision-makers to effectively assess the inclusiveness and performance of digital tools. Upon opening the dashboard, users are presented with an intuitive main interface that allows for easy navigation and a clear starting point for deeper analysis. Below is a breakdown of the main sections within the dashboard.

Main Interface

The main interface offers a user-friendly, structured design that emphasizes accessibility and streamlined navigation. Key features include:

Header and Navigation Bar: Located at the top, this bar allows users to filter and customize views by tool type, country, portfolio, and technology, aligning data with user priorities.

Summary Cards: Positioned on the right, these cards present key metrics such as user demographics and inclusiveness scores, offering a quick overview of essential data.

Informational Text Boxes: Definitions of "Digital Inclusiveness" and "MDII" provide brief context, helping users understand the dashboard's focus on assessing digital tool inclusiveness.

Dashboard Overview

When users open the MDII Scoring Dashboard, they are presented with an initial overview that highlights key performance metrics and the primary inclusiveness dimensions for each digital tool, as illustrated in Figure 3. This section serves as a starting point for further exploration, with insights into each tool's inclusiveness and performance metrics.

General Overview Section

As shown in Figure 3, the upper part of the dashboard delivers a broad overview of each digital tool's performance across seven primary dimensions: Accessibility, Beneficial Impact, Usage Effectiveness, Ethical & Responsible Innovation, Co-Creation & Governance, Supportive Ecosystem, and Risks & Harms.

This section enables users to assess key data points like the end user count and system type for each tool, categorizing them by focus area within the aFWL sectors. It also includes aggregate metrics, such as the total number of digital tools and the number of countries where these tools are deployed, providing a high-level understanding of the portfolio's geographic reach and overall scope.

MDII Score Breakdown

This section provides a focused analysis of the MDII score (Martins *et al.*, 2023) across the seven dimensions as shown in Figure 4, helping users gain a deeper understanding of overall inclusiveness. A summary card prominently displays the average MDII score, while a radar chart visualizes the distribution of scores across the dimensions, revealing

Figure 3 :General overview of digital tools

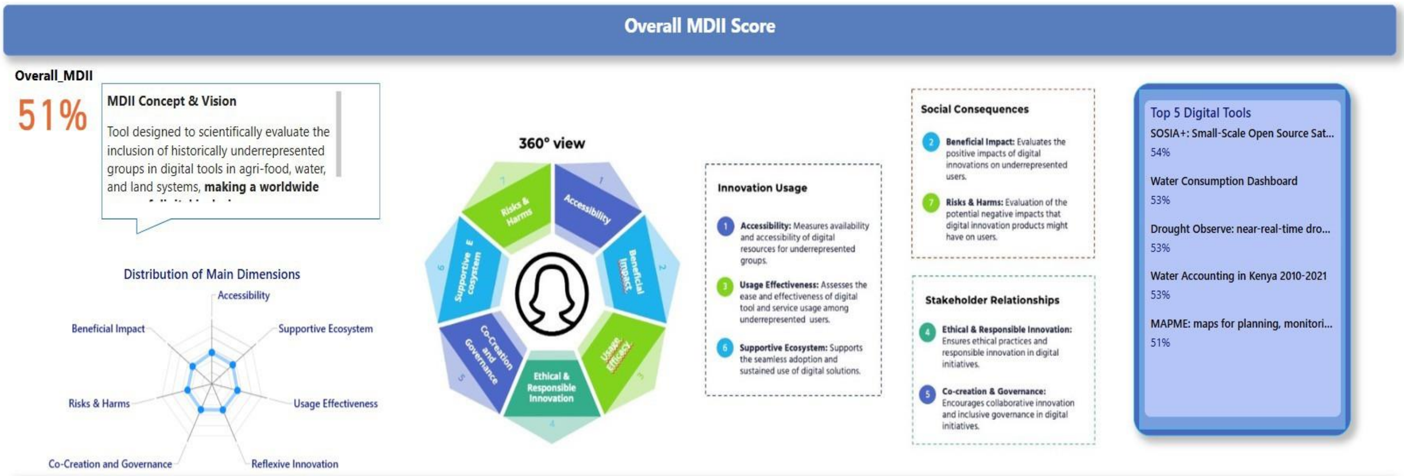

Figure 4: Overall MDII score section

strengths and imbalances. The radar chart and score card together allow users to assess how the portfolio performs holistically, offering insight into the degree of inclusiveness achieved across various aspects of digital engagement.

Detailed Dimension Section

In addition to the overall section, it becomes possible to have in-depth comparison of digital tools across the various domains, as well as with each sub-dimension's performance presented through a well-structured layout (see figure 5):

Sub-Dimension Distribution: On the left side of each sub-dimension section, a visual representation of the numerical distribution is displayed. This helps identify how digital tools are performing across the specific sub-dimension, making it easy to spot strengths and areas for improvement.

Metadata Overview: At the center of the section, key metadata related to the sub-dimension is provided, offering contextual information about each tool's performance in that area. This ensures a comprehensive understanding of the factors influencing the performance.

Voice of Beneficiaries: Qualitative data, such as feedback and experiences from end users, is featured in a dedicated section. This provides insight into how the tools are impacting beneficiaries, helping to connect quantitative metrics with real-world outcomes.

Top 5 Digital Tools: On the right side, the top five highest-scoring digital tools for each dimension are highlighted. This allows for quick identification of the best-performing tools and facilitates benchmarking across the portfolio.

Figure 5: Components of each Domain, Accessibility Dimension example.

USING DASHBOARD FEATURES

Filtering and customizing data views

The dashboard includes customizable filtering features, allowing users to easily explore specific innovations and geographic areas. With filters for Digital Tools, Country, Portfolio, and Technology, users can tailor their analysis to focus on particular tools or regions, enhancing the relevance of their insights.

Digital Tool filter: Selects individual digital tools to view specific performance metrics and inclusiveness scores.

Country filter: Analyzes the geographic distribution of digital tools, highlighting their presence in different regions.

Portfolio filter: Concentrates on specific collections of tools for comparative analysis in defined Context.

Technology filter: Examines tools based on employed technologies, identifying trends related to specific technical approaches.

Tracking Geographic Reach and Impact

The dashboard includes features for visualizing the geographic impact of digital tools:

Geographical Spread Analysis: The map visualization shows the regions where digital tools are deployed, providing insights into their geographic reach and impact. Figure 7 illustrates this visualization, highlighting areas of deployment and allowing users to easily assess the geographic scope and influence of each tool.

Figure 6: Filters by Digital Tool, Country, Portfolio and Technology used by the Digital Tool

Figure 7: Geographic Distribution of Digital Tools

Analyzing Digital Tools' Performance

The overall performance of digital tools can be quickly assessed through the following features:

System Type Analysis: The dashboard displays whether a digital tool is focused on food, water, or land systems by showing system types. For example, the gauge chart in Figure 8 demonstrates that 100% of the digital tools currently displayed are focused on water systems. This visualization allows users to easily identify the primary system type targeted by the tools, offering a straightforward method of evaluating their sectoral focus, as well as the distribution of tools across systems.

Figure 8: The gauge chart of system type analysis

Viewing Key Metrics: The dashboard provides a clear view of important metrics, such as the total number of end users or inclusiveness scores. For instance, Figure 9 shows the gender distribution of tool users, highlighting the number of females and males using the tool. This insight is critical for innovators who have developed a tool specifically for female users, as they can easily assess whether their target demographic is being reached and make informed decisions based on the user count.

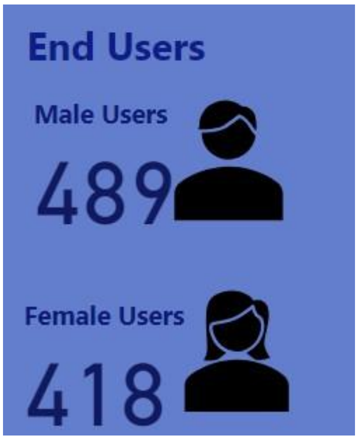

Figure 9: Cards of count of end users

Exploring Digital Innovation Applications

Detailed insights are important because they provide context and clarity on the specific applications and effectiveness of digital tools, allowing portfolio managers and decision-makers to determine alignment with strategic goals and policy priorities. These insights help to identify gaps or redundancies, ensuring resources are allocated efficiently and highlighting areas for potential improvement or rebalancing within key sectors. Detailed insights into the applications of digital tools are available through:

Digital Innovation Breakdown: A detailed analysis of digital tools across different sectors offers valuable insights into their specific use cases. This section categorizes tools by application areas such as hydrology, consumption, and resource management, providing a clearer understanding of their primary functions. This breakdown enables project managers and decision-makers to assess whether the tools

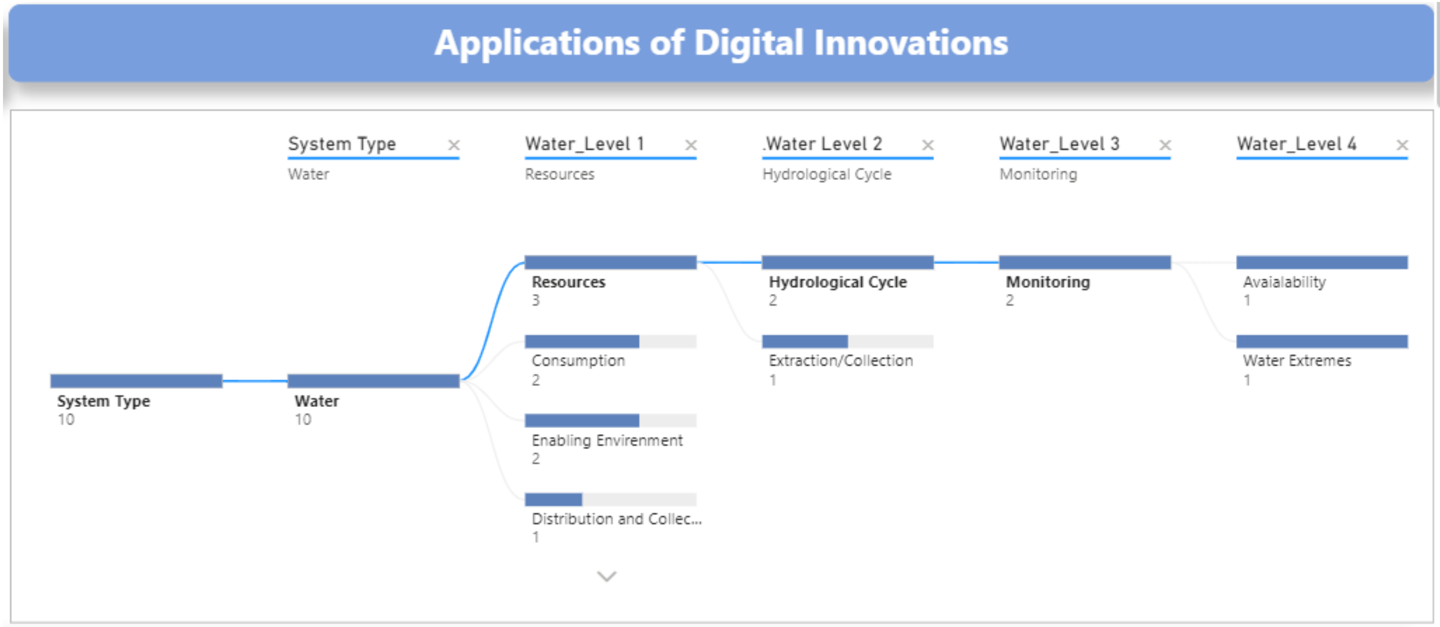

Figure 10: Tree Map of Applications of Digital Innovations

align with strategic goals and policies. It also helps identify potential surpluses, deficits, or balanced distributions of tools within key areas of the policy agenda.

Understanding the Context with Rationale Notes

Rationale notes accompany each visualization:

Rationale for Visualizations: Explanatory notes provide context for each visualization, clarifying the decisions behind the data presentation and the rationale.

Engaging Visual Elements

The dashboard features a range of interactive visualizations, including:

Gauge charts to indicate the system type, such as whether the digital tool focuses on food, water, or land.

Cards for displaying key metrics at a glance, helping users quickly assess important data points.

Decomposition trees to break down and display the different applications of digital innovations.

Maps for geographic analysis, showing the reach and impact of digital tools across regions.

Radar charts to visualize performance across multiple sub-dimensions simultaneously, providing a comparative overview.

Waffle charts to represent qualitative data, such as the voice of beneficiaries, offering insight into end-user feedback.

DETAILED DIMENSION SECTION

The MDII Scoring Dashboard offers a detailed analysis of digital tools' performance across seven dimensions of digital inclusiveness. Each dimension reflects a critical aspect of inclusiveness, incorporating both quantitative data and qualitative insights. The dashboard is structured to enable detailed exploration of each dimension to better understand how digital tools perform across various sub-categories.

Accessibility

This dimension measures availability and accessibility of digital resources for marginalized groups. The accessibility dimension forms the basis of digital engagement for marginalized users, aiming to bridge the digital divide by ensuring foundational access to digital resources.

As shown in Figure 5, this section highlights the distribution of accessibility sub-dimensions – Digital Accessibility, Economic Accessibility, Infrastructure Accessibility, and Information Accessibility – providing insights into areas that may need further focus to enhance digital tool accessibility. Figure 9 shows the metadata, which includes tags such as Climate, Environment, Gender, Poverty, and Nutrition, scored from 0 (Not Targeted) to 2 (Principal), visually representing each tool's alignment with CGIAR strategy objectives. Voice of Beneficiaries captures qualitative feedback with the statement, "I feel the digital tool focuses on problems that really matter to underrepresented groups and communities," offering insights into device compatibility, ease of use, and access barriers, thereby connecting tool performance to real-world accessibility needs. Finally, the Top 5 Digital Tools list highlights the highest-scoring tools, making it easy to identify those excelling in accessibility across sub-dimensions.

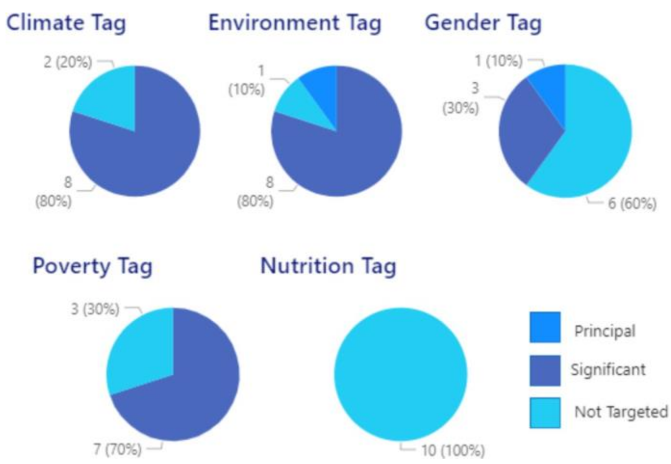

Figure 11: Meta Data in Accessibility Dimension: Distribution of Tags in Pie charts

Beneficial Impact

This dimension evaluates the positive impacts of digital innovations on marginalized users. This dimension is significant as it measures the real-world benefits and positive impacts of digital solutions in improving the quality of life and addressing the unique challenges faced by marginalized users.

As shown in Figure 12, the distribution of Beneficial Impact sub-dimensions – including Problem Relevance, Sustainability, Social Value Creation, and Solution Effectiveness – enables a focused analysis of where digital tools excel and identifies areas needing improvement. Metadata on Impact Areas (e.g., Climate Adaptation and Mitigation) and Organization Types (e.g., NGOs, government entities, private sector) offers insights into the sectors and purposes these tools serve. The Voice of Beneficiaries section captures user feedback on inclusiveness through responses to questions like, "I feel the digital tool focuses on problems faced by underrepresented groups," providing critical insights from end users about the tool’s relevance to their needs.

USAGE EFFECTIVENESS

This dimension assesses the ease and effectiveness of digital tool and service usage among marginalized users. The Usage Efficacy dimension is important to evaluate how effectively

Figure 12: The section of Beneficial Impact Dimension

digital solutions meet the unique needs and circumstances of marginalized users, promoting ease of use and beneficial outcomes, thereby facilitating digital inclusion.

The distribution of Usage Effectiveness sub-dimensions—including Usability, Desirability, and Digital Skill Enhancement—provides insights into the accessibility and practical value of digital tools for various communities, helping to pinpoint areas that may need more focus. Metadata on user demographics, such as Elder Users, Youth Users, Unschooled Users, and Women Users, reveals the specific groups engaging with these tools and the extent of their digital skill enhancement. The Voice of Beneficiaries section captures feedback on statements like, "I think this tool and its benefits are properly communicated to me or our community," providing qualitative insights into the clarity and effectiveness of the tools' purpose for users. Lastly, the Top 5 Digital Tools section highlights the tools that perform best in usage, engagement, and skill-building within target communities.

Ethical & Responsible Innovation

This dimension ensures ethical practices and responsible innovation in digital initiatives. The ethical and responsible innovation enables greater reassurance that digital innovations are ethically sound and responsibly managed, promoting justice, fairness and responsible innovation for marginalized users.

The Distribution section visually breaks down sub-dimensions of Reflexive Innovation, Data Governance, and Ethical Compliance, allowing users to analyze areas within ethical standards that may require additional focus, such as data governance, ethical compliance, or promoting reflexive innovation. Metadata on Technology Used offers insights into each tool's technological framework, which may influence its ethical and responsible innovation practices, adding context to its ethical performance. The Voice of Beneficiaries section includes user feedback on the statement, "I feel I will be offered support to enable me to collaborate in developing the tool," providing qualitative insights into the tools' inclusivity and potential for user collaboration.

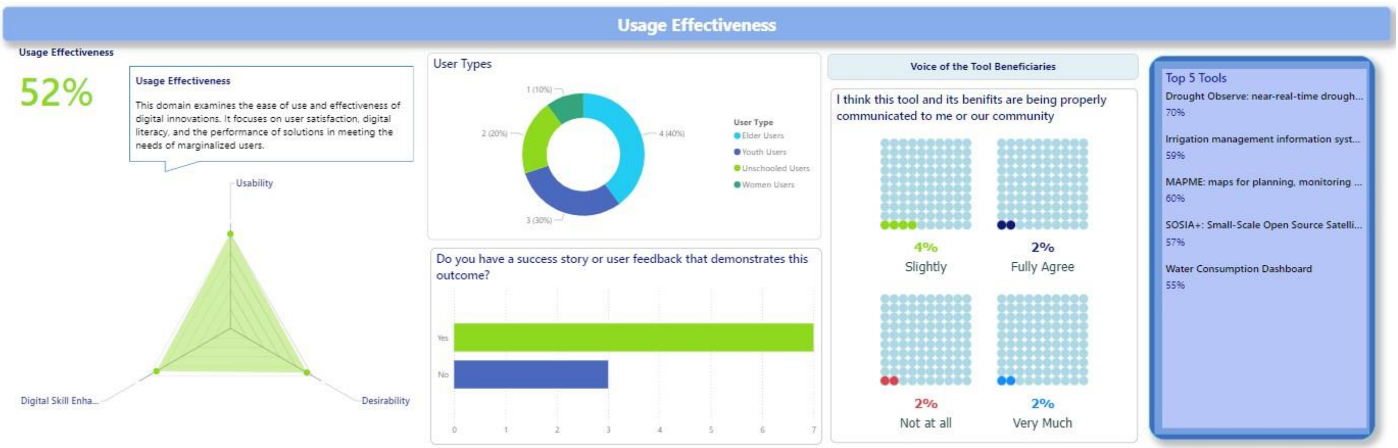

Figure 13: The section of Usage Effectiveness Dimension

Figure 14: The section of Ethical and Responsible Innovation

Co-Creation and Governance

This dimension encourages collaborative innovation and inclusive governance in digital initiatives. Co-creation and governance are important as high-level assessment for inclusive and collaborative design, ensuring that digital solutions are cocreated with marginalized users, governed inclusively, and are accessible, and effective dispute resolution processes for marginalized users, promoting justice and resolution.

The Distribution section features a radar chart that visualizes sub-dimensions such as Inclusive Governance, Grassroots Innovation Collaboration, and Collaborative Innovation, allowing users to identify strengths and areas for improvement in collaborative efforts, ensuring balanced stakeholder involvement. The Metadata section provides information on the target sub-groups reached by the digital tools, including Elder Users, Unschooled Users, and Women Users, highlighting the inclusivity of the innovation process. In the Voice of Beneficiaries section, user feedback is captured through the question, "I feel I will be offered support to enable me to collaborate in developing the tool," offering qualitative insights into the inclusiveness of the co-creation process.

Risks and Harms

This dimension assesses the potential negative impacts of digital innovations on users, especially underrepresented groups within agrifood systems. By addressing risks and ensuring that solutions are safe, effective, and socially responsible, it highlights the importance of protecting vulnerable populations from unintended consequences.

The Distribution section features a radar chart that breaks down sub-dimensions such as Data Misappropriation, Long- Term Loss, Lasting Inequalities, and Reinforcement of Inequalities. These sub-dimensions help users identify areas where digital tools may inadvertently cause harm, providing clarity on where improvements are needed to safeguard vulnerable communities. The Metadata section includes information on the duration of digital tool usage and whether concrete actions have been taken to address Gender Equality and Social Inclusivity (GESI) during the tool's development.

Figure 15: The section of Risks and Harms Dimension

This information aids in assessing the tool's long-term impact and commitment to inclusivity. In the Voice of Beneficiaries section, feedback is captured through the question, "I think this tool and its benefits were properly communicated to me or our community," offering qualitative insights into the effectiveness of the tool's communication and management of potential risks.

Supportive Ecosystem

This dimension evaluates the level of community engagement, the presence of enabling policies, and the availability of resources that support underrepresented communities in engaging with digital tools. It highlights the broader ecosystem necessary for fostering digital inclusiveness and sustainable engagement. The Distribution section features a radar chart that divides the Supportive Ecosystem into sub-dimensions, including Community Engagement, Policy Support, and Resource Availability. These sub-dimensions provide insight into the strength of the ecosystem surrounding digital tools, helping to identify areas that may require additional support to foster inclusivity. The Metadata section focuses on policies and initiatives backing the digital tools, alongside resources available to users, such as training and technical support. This information clarifies the external factors contributing to the successful adoption and sustainability of digital tools. In the Voice of Beneficiaries section, user feedback is captured through the question, "I feel supported by the existing policies and resources when engaging with this digital tool." This qualitative input helps gauge whether the supportive ecosystem is effectively addressing the needs of underrepresented communities.

TECHNICAL IMPLEMENTATION

Deployment Platform

Power BI: The dashboard is deployed on Power BI, utilizing its strengths in real-time data processing, interactive visualizations, and seamless connection to data sources. The platform is integrated with CGIAR accounts, ensuring secure access and scalability, with the flexibility to replace dummy data with live data as it becomes available.

Data Sources and Integration Process

Dummy Data & Python Integration: In this initial version, dummy data was created to simulate real-world metrics, with MDII scores extracted through Python scripts. This data forms the basis for visualizations and interactions in the dashboard.

Data Types:

The dashboard supports both quantitative (e.g., performance metrics, usage statistics) and qualitative (e.g., feedback) data inputs.

Quantitative Data: This type of data forms the core of the dashboard's scoring system. It includes:

MDII Index Values and Dimension Scores: These numerical values provide a standardized measurement of digital inclusiveness across the seven key dimensions: Accessibility, Beneficial Impact, Usage Effectiveness, Ethical & Responsible Innovation, Co-Creation and Governance, Supportive Ecosystem, and Risks & Harms.

Metadata of Digital Tools: Information about the tools, such as demographic reach, and geographic focus, technology used by the digital tool is quantified and presented to help users understand the performance and impact of each tool.

Performance Metrics: Key performance indicators (KPIs) related to the effectiveness, reach, and engagement of digital tools are included to provide clear, measurable insights. This encompasses various KPIs, such as the number of end users, which indicates the tool's reach and adoption.

Qualitative Data: In addition to the quantitative metrics, qualitative insights are incorporated to capture the voice and experiences of end users. These include:

Voice of Beneficiaries (End Users): Direct feedback from the users of digital tools, reflecting their experiences, challenges, and satisfaction. This provides crucial context to the quantitative scores, offering a more nuanced understanding of tool effectiveness and inclusiveness.

CONCLUSION

The MDII Scoring Dashboard is a vital tool for promoting digital inclusiveness in agri-food systems by allowing data to tell a story. By bringing together various dimensions and incorporating valuable feedback from the marginalized users and unrepresented communities, it provides a clear picture of how digital tools are performing and where they may fall short in serving underrepresented groups. This dashboard not only showcases successful strategies but also encourages critical thinking about areas that need more focus, allowing for more informed decision-making.

Looking forward, the next steps involve integrating real data into the dashboard, which will make it even more relevant and precise. The insights gained from this process will empower users to pinpoint specific dimensions where their tools might not be meeting expectations, paving the way for targeted improvements. For instance, if a particular dimension scores lower than desired, users can delve into the reasons behind this and work towards developing more effective digital solutions.

In summary, the MDII Scoring Dashboard marks a significant step toward creating more inclusive digital environments within agrifood systems. By fostering collaboration and upholding ethical practices, it aligns with the broader mission of empowering marginalized communities and ensuring everyone has equitable access to digital innovations.

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